

Physics ;

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Assignment - 1

Q1. Explain what it means by overclocking PC?

→ Overclocking a gaming PC means forcing your computer components to run faster than the speed limit set by the manufacturer. By increasing the operating frequency your hardware processes data quicker. This translates directly into higher frame rate (FPS) and smoother gameplay without buying new parts.

Q2. A  $12\ \Omega$  resistor is connected to a 24 V supply, find the flowing current through the resistor.

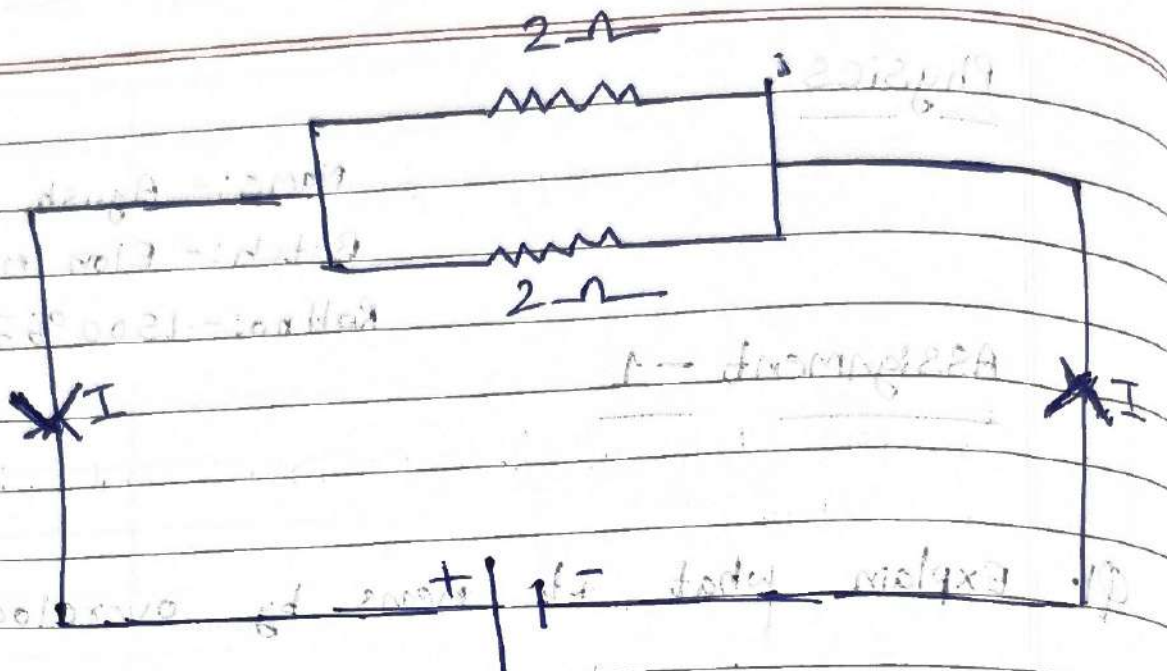
→  $V = IR$  — (1)

$$I = \frac{V}{R} \quad \text{--- (2)}$$

$$= \frac{24}{12}$$

$$I = 2A$$

Q3.



Find the current?

→ Resistance in parallel =  $\frac{R_1 R_2}{R_1 + R_2}$

$$\therefore R = \frac{2 \times 2}{2 + 2} = 1 \Omega$$

$$R = 1 \Omega$$

$$V = IR$$

$$\therefore I = \frac{V}{R}$$

$$I = 5 A$$

Q4. A resistor of  $6\ \Omega$  is connected to  $12\text{ V}$  battery, a charge particle of mass  $2\text{ kg}$ , experiences a force equal to the weight of current flowing through the resistor. Find its acceleration.

→ Given:  $R = 6\ \Omega$   
 $V = 12\text{ V}$   
 $M = 2\text{ kg}$   
 $I = ?$

To find:  $a = ?$

Sol:

$$V = IR$$

$$I = \frac{V}{R} = \frac{12}{6}$$

$$I = 2\text{ A}$$

$$I = F$$

$$F = ma$$

$$a = \frac{F}{m}$$

$$a = 1\text{ m/s}^2$$

Q5. What is Electromagnetism?

→ The concept:-

- i] When electricity flows, it creates a magnetic field. Magnetic fields can also produce electricity.
- ii] This connection helps move, control, and store power across the motherboard. It also allows parts like coils, inductors and tiny motors to work properly.

Examples:-

- i] Voltage regulator modules (VRM) use inductors to stabilize CPU power.
- ii] Cooling fan motors spin using electromagnetic coils.
- iii] Wireless charging pads transfer energy via electromagnetic induction.

Q6. i) Which of the following correctly describes the magnetic field near a long straight wire?

a) The field consist of straight line perpendicular to wire.

b) The field consist of straight lines parallel to wire.

c) The field consist of radial lines originating from wire.

☒ d) The field consist of concentric circles center of wire.

Q7. At the time of shortcircuit the current in circuit \_\_\_\_\_?

a) reduces substantially

b) Does not change.

c) increases heavily

☒ d) Vary continuously